

Johansen Cointegration Test

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Sample: 2011M01 2025M12

Included observations: 180

Lags interval (in first differences): 1 to 2

Endogenous variables: LNCPI LN_CIF LN_CUSTOMS_DUTIES_TAXES LN_OPE

Exogenous variables (short-run only)*: M2 M3 M4 M5 M6 M7 M8 M9 M10 M11 M12

Deterministic assumptions: Case 3 (Johansen-Hendry-Juselius): Cointegrating relationship includes a constant. Short-run dynamics include a constant.

* Warning: Critical values assume no exogenous series.

Unrestricted Cointegration Rank Test (Trace)

Hypothesized No. of CE(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.** Critical Value
None *	0.193072	107.5747	95.75366	0.0060
At most 1	0.182294	69.60444	69.81889	0.0520
At most 2	0.084904	33.98287	47.85613	0.5028
At most 3	0.058824	18.27823	29.79707	0.5456
At most 4	0.039671	7.547515	15.49471	0.5150
At most 5	0.002159	0.382591	3.841465	0.5362

Trace test indicates 1 cointegrating equation(s) at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegration Rank Test (Max-eigenvalue)

Hypothesized No. of CE(s)	Eigenvalue	Max-Eigen Statistic	0.05 Critical Value	Prob.** Critical Value
None	0.193072	37.97027	40.07757	0.0847
At most 1 *	0.182294	35.62157	33.87687	0.0307
At most 2	0.084904	15.70464	27.58434	0.6907
At most 3	0.058824	10.73071	21.13162	0.6741
At most 4	0.039671	7.164924	14.26460	0.4698
At most 5	0.002159	0.382591	3.841465	0.5362

Max-eigenvalue test indicates no cointegration at the 0.05 level

Unrestricted Cointegrating Coefficients (normalized by b*S11*b=I):

LNCPI	LN_CIF	LN_CUSTO...	LN_OPEX	IR	LN_FX
14.72580	-2.397516	-5.995405	-2.917690	-0.051030	31.46081
78.69269	3.031709	-6.549726	-0.143074	0.942854	-5.721208
142.2928	1.455785	-6.325171	-0.271148	-1.559351	9.586127
40.27396	-0.502433	-5.285478	1.070727	-0.477571	14.34188
13.59092	-2.619601	4.078871	-1.404556	-0.480246	-12.58675
159.6097	-0.159236	-4.613941	1.046309	0.101496	-0.839025

Unrestricted Adjustment Coefficients (alpha):

D(LNCPI)	0.000177	2.45E-05	4.10E-06	-0.000108	-4.05E-05	3.92E-06
D(LN_CIF)	0.035973	-0.018860	-0.005807	0.013877	0.020554	0.004000
D(LN_CUSTO...	0.015508	0.012391	0.021509	0.004448	0.002073	6.29E-05
D(LN_OPEX)	0.018929	-0.020710	0.013761	-0.018070	0.025134	-0.005108
D(IR)	0.006935	-0.080631	0.016865	0.012479	-0.012968	-0.001605
D(LN_FX)	-0.003141	-0.000666	0.002151	-0.001435	0.000648	0.000339

1 Cointegrating Equation

Log-Likelihood: 1869.337

Normalized cointegrating coefficients (standard error in parentheses)

LNCPI	LN_CIF	LN_CUSTO...	LN_OPEX	IR	LN_FX
1.000000	-0.162811	-0.407136	-0.198135	-0.003465	2.136441
	(0.05304)	(0.09954)	(0.04004)	(0.02055)	(0.42564)

Adjustment coefficients (standard error in parentheses)

D(LNCPI)	0.002611 (0.00074)
D(LN_CIF)	0.529733 (0.20371)
D(LN_CUSTO...	0.228361 (0.10510)
D(LN_OPEX)	0.278746 (0.23780)
D(IR)	0.102127 (0.25976)
D(LN_FX)	-0.046259 (0.01677)

2 Cointegrating Equations

Log-Likelihood: 1887.147

Normalized cointegrating coefficients (standard error in parentheses)

LNCPI	LN_CIF	LN_CUSTO...	LN_OPEX	IR	LN_FX
1.000000	0.000000	-0.145211	-0.039383	0.009026	0.350019
		(0.02432)	(0.00855)	(0.00474)	(0.09881)
0.000000	1.000000	1.608771	0.975066	0.076721	-10.97240
		(0.49732)	(0.17481)	(0.09700)	(2.02084)

Adjustment coefficients (standard error in parentheses)

D(LNCPI)	0.004539	-0.000351
	(0.00402)	(0.00019)
D(LN_CIF)	-0.954445	-0.143425
	(1.10435)	(0.05332)
D(LN_CUSTO...)	1.203457	0.000387
	(0.56758)	(0.02740)
D(LN_OPEX)	-1.350955	-0.108168
	(1.29005)	(0.06228)
D(IR)	-6.242980	-0.261078
	(1.31589)	(0.06353)
D(LN_FX)	-0.098703	0.005511
	(0.09135)	(0.00441)

3 Cointegrating Equations
Log-Likelihood: 1895.000

Normalized cointegrating coefficients (standard error in parentheses)					
LNCPI	LN_CIF	LN_CUSTO...	LN_OPEX	IR	LN_FX
1.000000	0.000000	0.000000	0.007990 (0.01003)	-0.026751 (0.00608)	0.056510 (0.05043)
0.000000	1.000000	0.000000	0.450220 (0.17937)	0.473081 (0.10867)	-7.720663 (0.90169)
0.000000	0.000000	1.000000	0.326240 (0.10240)	-0.246374 (0.06204)	-2.021255 (0.51475)
Adjustment coefficients (standard error in parentheses)					
D(LNCPI)	0.005122 (0.00823)	-0.000345 (0.00021)	-0.001249 (0.00055)		
D(LN_CIF)	-1.780754 (2.25834)	-0.151879 (0.05713)	-0.055412 (0.15080)		
D(LN_CUSTO...)	4.264089 (1.12539)	0.031700 (0.02847)	-0.310183 (0.07515)		
D(LN_OPEX)	0.607119 (2.63324)	-0.088136 (0.06661)	-0.064884 (0.17583)		
D(IR)	-3.843240 (2.68310)	-0.236527 (0.06787)	0.379862 (0.17916)		
D(LN_FX)	0.207347 (0.18471)	0.008642 (0.00467)	0.009594 (0.01233)		

4 Cointegrating Equations
Log-Likelihood: 1900.365

Normalized cointegrating coefficients (standard error in parentheses)					
LNCPI	LN_CIF	LN_CUSTO...	LN_OPEX	IR	LN_FX
1.000000	0.000000	0.000000	0.000000	-0.025375 (0.00633)	0.063899 (0.05247)
0.000000	1.000000	0.000000	0.000000	0.550590 (0.13958)	-7.304308 (1.15750)
0.000000	0.000000	1.000000	0.000000	-0.190210 (0.08661)	-1.719555 (0.71820)
0.000000	0.000000	0.000000	1.000000	-0.172158 (0.21390)	-0.924780 (1.77380)
Adjustment coefficients (standard error in parentheses)					
D(LNCPI)	0.000773 (0.00837)	-0.000291 (0.00021)	-0.000679 (0.00060)	-0.000638 (0.00016)	
D(LN_CIF)	-1.221854 (2.32598)	-0.158852 (0.05755)	-0.128761 (0.16758)	-0.085826 (0.04320)	
D(LN_CUSTO...)	4.443246 (1.16139)	0.029465 (0.02874)	-0.333696 (0.08367)	-0.048088 (0.02157)	
D(LN_OPEX)	-0.120628 (2.70985)	-0.079057 (0.06705)	0.030624 (0.19524)	-0.075345 (0.05033)	
D(IR)	-3.340647 (2.76744)	-0.242797 (0.06847)	0.313902 (0.19939)	9.05E-05 (0.05140)	
D(LN_FX)	0.149568 (0.18986)	0.009363 (0.00470)	0.017177 (0.01368)	0.007141 (0.00353)	